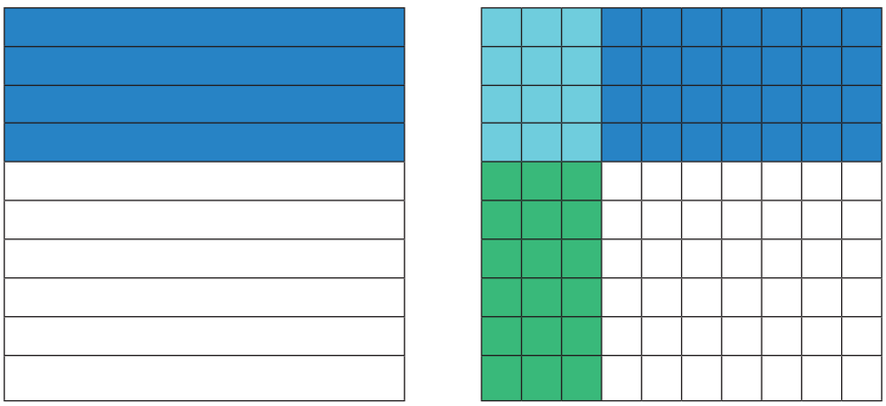
Example 1.16 Find 0.3×0.4

Solution

First shade 4 rows of the grid in blue colour to represent 0.4. Shade 3 columns of the grid in green colour to represent 0.3 of 0.4. Now 12 squares represents the common portion. This represents 12 hundredth or 0.12. Hence $0.3 \times 0.4 = 0.12$.

$\frac{4}{10}$ $\qquad$ $\frac{3}{10}^{\text{th}}$ of $\frac{4}{10} = \frac{12}{100}$



Note

The number of decimal digits in 0.12 is two. So, we can conclude that the number of decimal digits in the product of two decimal numbers is equal to the sum of decimal digits that are multiplied.

Area model

We have already learnt about the area model in the addition and subtraction of decimal numbers. In the same way we are going to multiply the decimal numbers. Now we shall see an example.

Example 1.17 Multiply 3.2×2.1

Solution

Let us try to represent the product of decimal numbers (3.2×2.1) as the area of a rectangle. Let us consider a rectangular portion as shown in *Fig. 1.9*.



Fig. 1.9

The rectangular portion is split into 3 wholes and 2 tenth along it's length (*Fig. 1.10*).

Q5Z2I3

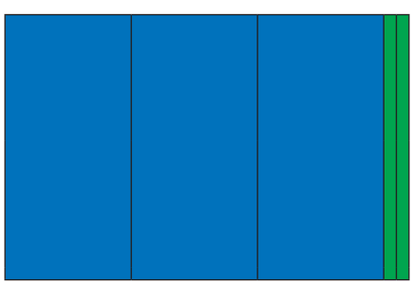


Fig. 1.10

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